

1. Which stock do you believe provides the most attractive investment

opportunity for the Cavalier Fund? Why?

Based on anticipated returns and risk metrics, **Delphi Automotive** appears to be the most attractive investment opportunity for the Cavalier Fund. Delphi offers a strong expected return of **9.1%**, paired with a **moderate risk profile**—its annualized standard deviation of **24.8%** is significantly lower than that of **Groupon (67.3%)** and **Kinross Gold (65%)**, and only moderately higher than **Kellogg (14.6%)**.

Using the expected returns, historical standard deviations, and a **risk-free rate of 3.4%**, Delphi has the **highest Sharpe Ratio** among the four stocks. This indicates that Delphi delivers the **best return per unit of risk**, making it the most efficient stock on a risk-adjusted basis. Its beta of **1.30** suggests that while it has slightly higher systematic risk than the overall market, the level of risk remains within a reasonable range.

From a valuation standpoint, Delphi's **P/E ratio of 13.6** and **Baa2 bond rating** reflect solid financial health and reasonable pricing. Additionally, its **dividend yield of 1.52%** enhances total return potential for long-term investors.

Strategically, Delphi operates in the **automotive parts industry**, which stands to benefit from trends such as **vehicle electrification and automation**. These industry tailwinds position Delphi for sustained growth, adding to its investment appeal.

Given the Cavalier Fund's mandate to **outperform the S&P 500** by selecting **undervalued, high-potential stocks**, Delphi Automotive offers the **best balance of return, risk, and strategic opportunity**, making it the most suitable investment among the four options.

2. How would you characterise the riskiness of the different stocks being considered by the Cavalier Fund? Which stock do you believe is the riskiest, and which is the least risky?

The riskiness of each stock under consideration by the Cavalier Fund can be evaluated using two key metrics: **standard deviation of returns** (which captures total volatility) and **beta** (which measures market or systematic risk). These metrics provide insight into both the total and market-related uncertainty associated with each investment.

Based on the data provided

Groupon is the **riskiest stock** in terms of both **total volatility** and **systematic risk**. It has the **highest standard deviation (67.3%)** and a **high beta (1.44)**, indicating that its returns are not only highly volatile but also very sensitive to overall market movements. This aligns with concerns raised by the team about Groupon's unpredictable return behavior.

Kinross Gold, while having slightly lower volatility than Groupon, still shows a **very high standard deviation (65.0%)**, suggesting substantial total risk. However, its **low beta (0.31)** indicates that most of this risk is idiosyncratic rather than market-driven—likely due to commodity price swings and geopolitical factors.

Kellogg stands out as the **least risky stock**. It has the **lowest standard deviation (14.6%)** and a **low beta (0.54)**, reflecting its nature as a stable, consumer-focused company with consistent cash flows and modest sensitivity to market fluctuations.

In summary:

- **Most risky (total + market risk): Groupon**
- **Least risky (total + market risk): Kellogg**

Delphi lies in the middle, offering a reasonable risk-return trade-off with **moderate volatility** and a **beta of 1.30**, making it suitable for balanced risk strategies.

3. How does portfolio diversification affect your characterisation of risk for the stocks?

Diversification is a foundational principle of modern portfolio theory. It reduces overall portfolio risk by combining assets whose returns are **not perfectly correlated**. In the context of the Cavalier Fund case, this concept is particularly relevant when considering the inclusion of high-volatility stocks like **Groupon** and **Kinross Gold**.

Individually, both stocks are extremely risky: Groupon has a standard deviation of **67.3%** and Kinross Gold **65.0%**, making them the most volatile options among the four under consideration. However, when held together in a **50-50 portfolio**, the overall standard deviation is actually **lower than that of either stock alone**. This counterintuitive result stems from the fact that the **price movements of the two stocks are not perfectly synchronized**—their returns are likely weakly correlated or even negatively correlated.

When two assets have low or negative correlation, **negative returns in one stock can be partially offset by positive returns in the other**, reducing the overall variability in the portfolio. This effect is a direct consequence of diversification, which enables investors to **mitigate idiosyncratic (firm-specific) risk**, while leaving only **systematic (market-related) risk**, which cannot be diversified away.

Sam Kramer's argument highlights a more refined way of thinking about risk—not just looking at how risky a stock is on its own, but how it behaves as part of a portfolio. In the case of Kinross Gold, while the stock appears highly volatile individually, it can still improve the overall portfolio by being paired with the right asset, such as Groupon. This is because their returns are not perfectly correlated, and combining them can lower total volatility. Kramer's point shows that a stock's real impact comes from how it interacts with other

holdings, not just from its own risk level. It's the marginal contribution to overall portfolio risk that truly matters when making smart investment decisions.

In summary, diversification transforms our risk assessment—from evaluating absolute volatility to analysing a stock's **marginal contribution to total portfolio risk**, which is crucial for optimal fund construction.

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$$

4. Kramer argues that the returns associated with a 50-50 weighted portfolio of Groupon and Kinross Gold have a lower standard deviation than those associated with either Groupon or Kinross Gold stock alone. Is that true?

Yes, Kramer's argument is valid. Both **Groupon** and **Kinross Gold** are individually very risky, with standard deviations of **67.3%** and **65.0%** respectively. But when these two stocks are combined in a **50-50 portfolio**, the resulting standard deviation is actually **lower than either of them alone**.

This happens because their returns are **not perfectly correlated**. In fact, assuming a correlation of **-0.2**, the standard deviation of the portfolio comes out to be around **41.8%**. The reason for this drop in risk is that the negative or low correlation allows losses in one stock to be partially offset by gains in the other. In other words, they don't move in the same direction at the same time, which smooths out the overall volatility.

Kramer's point shows how diversification works in practice. Even though each stock is risky on its own, when combined, the **portfolio becomes less volatile**. This is a key idea in portfolio theory: it's not just the individual risk of a stock that matters, but how it behaves in combination with others.

So yes, the data supports Kramer's claim. The 50-50 portfolio of Groupon and Kinross Gold does have a **lower standard deviation**, proving that diversification can significantly reduce total portfolio risk—even when using two individually risky stocks.

5. What is the argument for Kinross Gold?

The main argument for including **Kinross Gold** in the Cavalier Fund is not based on its performance in isolation, but on how it improves the overall **risk-return profile** of the portfolio. On its own, Kinross is a **high-risk stock**, with a standard deviation of **65%**, and an expected return of **8.4%**. However, what makes it valuable is its **diversification benefit**.

Kinross Gold operates in the **gold mining industry**, which often behaves differently from the broader equity market. Because of this, its returns are **less correlated** with typical market movements. Its **low beta of 0.31** confirms that it carries little systematic (market) risk, and is

mostly influenced by factors unique to the gold sector. This makes it a useful hedge in a portfolio of mostly market-driven stocks.

Also, as seen in the analysis with **Groupon**, combining Kinross with another volatile but differently behaved stock can significantly reduce **overall portfolio volatility**. A 50-50 portfolio of Kinross and Groupon results in a **standard deviation of around 41.9%**, which is much lower than each stock individually.

So, while Kinross might not look attractive on its own, it plays an important **supporting role** in building a more stable and diversified portfolio. It allows the fund to take advantage of a decent expected return, while reducing risk through negative or low correlation with other holdings. That's the key argument for including it—it helps **manage risk without giving up return**, which aligns with the fund's long-term investment goals.

6. How should investors price risk? How should they trade off the relation between risk and return?

Investors should price risk based on how much return they expect for taking on that risk. The most common way to do this is through the **Capital Asset Pricing Model (CAPM)**. CAPM tells you what return a stock should give, based on its **beta**, or how much it moves with the overall market.

The formula is:

$$R_i = R_f + \beta_i (R_m - R_f)$$

Where:

R_i = Expected return on the stock

R_f = Risk-free rate

R_m = Market return

β = Stock's sensitivity to market (systematic risk)

This means a stock's return should be the risk-free rate plus a risk premium, which depends on how sensitive it is to market movements. Stocks with higher beta are riskier, so they should give higher returns to make that risk worth it.

The main idea is that investors should only be rewarded for **systematic risk** (market-related risk), not for **idiosyncratic risk** (firm-specific), because the latter can be reduced through diversification.

When making investment decisions, investors should compare the **expected return of a stock** to its **CAPM return**. If the expected return is **higher than CAPM**, the stock may be **undervalued** and a good buy. If it's lower, it might be **overvalued** for its level of risk.

So, the trade-off between risk and return is about finding the right balance—taking on more risk only when it's matched by more potential return. Investors shouldn't just chase high

returns; they should look for investments that offer the **best return for the risk taken**. This is what makes risk-adjusted measures like CAPM and the Sharpe Ratio useful in real decisions.

7. What is the risk-adjusted benchmark return on the S&P 500 portfolio? On the proposed stocks?

To find the risk-adjusted benchmark return, we use the **CAPM formula**:

$$R_i = R_f + \beta_i (R_m - R_f)$$

From the case:

- **Risk-free rate (R_f) = 3.4%**
- **Market return (R_m) = 12.0%** (arithmetic mean)
- **So, Market Risk Premium = 8.6%**

The returns for this calculated in the excel .

- For **Delphi**, with a beta of 1.30, the CAPM return is **14.58%**.
- For **Groupon**, with a beta of 1.44, the CAPM return is **15.78%**.
- For **Kellogg**, with a beta of 0.54, the CAPM return is **8.04%**.
- For **Kinross Gold**, with a beta of 0.31, the CAPM return is **6.07%**.
- The **S&P 500**, which has a beta of 1, gives a CAPM return of **12.0%**—this also serves as the fund's benchmark.

we compare them to the stocks' expected returns. If the expected return is **lower than the CAPM return**, the stock is **overvalued**—it's not offering enough return for the level of risk. If the expected return is **higher than the CAPM return**, the stock is **undervalued**—investors are being compensated more than expected for the risk

Using this approach, we find that **Delphi, Groupon, and Kellogg** all have expected returns **below their CAPM benchmark**, so they are **overvalued** on a risk-adjusted basis. On the other hand, **Kinross Gold** has an expected return that is **above its CAPM return**, which means it is **undervalued**.

So, based purely on CAPM, **Kinross Gold is the only stock that looks attractive** relative to its level of risk. The others are not offering enough return for the risk they carry.

8. How should Nickerson think about picking stocks for the Cavalier Fund?

Based on the analysis, **Delphi Automotive** and **Kinross Gold** are the best candidates for inclusion in the Cavalier Fund portfolio.

Delphi offers a solid expected return of **9.1%** with **moderate risk**, and has the **highest Sharpe ratio** among the four options. This means it provides the best return per unit of total risk. While its expected return is lower than the CAPM benchmark, Delphi still makes sense for the fund because of its balanced profile, strong fundamentals, and exposure to growth trends. It adds value through good return potential without excessive volatility.

Kinross Gold, on the other hand, is very risky on its own, but stands out in two ways. First, it's the **only stock undervalued** according to CAPM—its expected return is higher than what the model requires for its level of risk. Second, it offers **strong diversification benefits**. When combined with a high-volatility stock like Groupon, it lowers overall portfolio risk due to low or negative correlation. That makes Kinross a strategic addition to reduce risk at the portfolio level.

Nickerson should focus on stocks that **beat their CAPM benchmark** or provide **diversification benefits** that improve the fund's risk-return profile. He should avoid high-risk stocks unless they offer either **a strong return premium or reduce overall volatility**. Every stock should be backed by **bottom-up fundamental analysis**, but also checked against its **required return for the risk taken**.

The goal is to keep the portfolio **well-diversified**, and aligned with the fund's mandate to **outperform the S&P 500** on a risk-adjusted basis.